

AI & Machine Learning Internship

Task 1 Report

House Price Prediction using Linear Regression

Submitted By:

NIBSHAN JOVIN JOSEPH

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Organization:

Maincrafts

1. Introduction

Objective

The objective of this project is to develop a Linear Regression model to predict house prices using the California Housing dataset. The project demonstrates the complete machine learning workflow, including data exploration, visualization, preprocessing, model training, evaluation, and model saving.

2. Dataset Description

The California Housing dataset was obtained from the Scikit-learn library. It contains information collected from different housing districts in California.

The dataset includes features such as:

- Median Income

- House Age
- Average Rooms
- Average Bedrooms
- Population
- Average Occupancy
- Latitude
- Longitude

The target variable is the median house value.

3. Tools and Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - Joblib
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4. Project Workflow

The following steps were completed during the project:

- Imported the required libraries.
- Loaded the California Housing dataset.
- Explored the dataset.
- Checked for missing and duplicate values.
- Visualized the data using different plots.

- Prepared the dataset for training.
 - Split the dataset into training and testing sets.
 - Trained a Linear Regression model.
 - Predicted house prices.
 - Evaluated the model using regression metrics.
 - Saved the trained model.
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5. Data Exploration

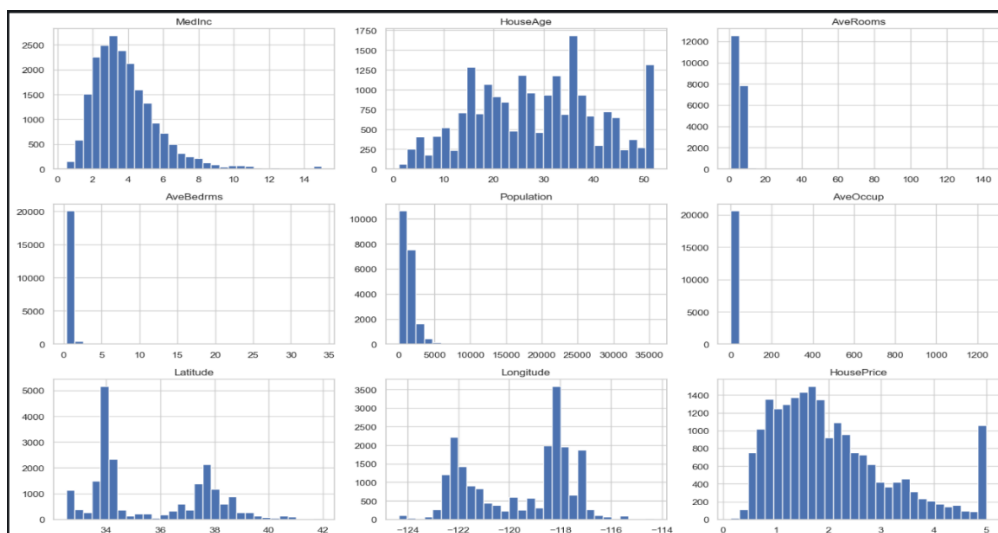
The dataset was explored using statistical analysis and visualizations.

The following visualizations were created:

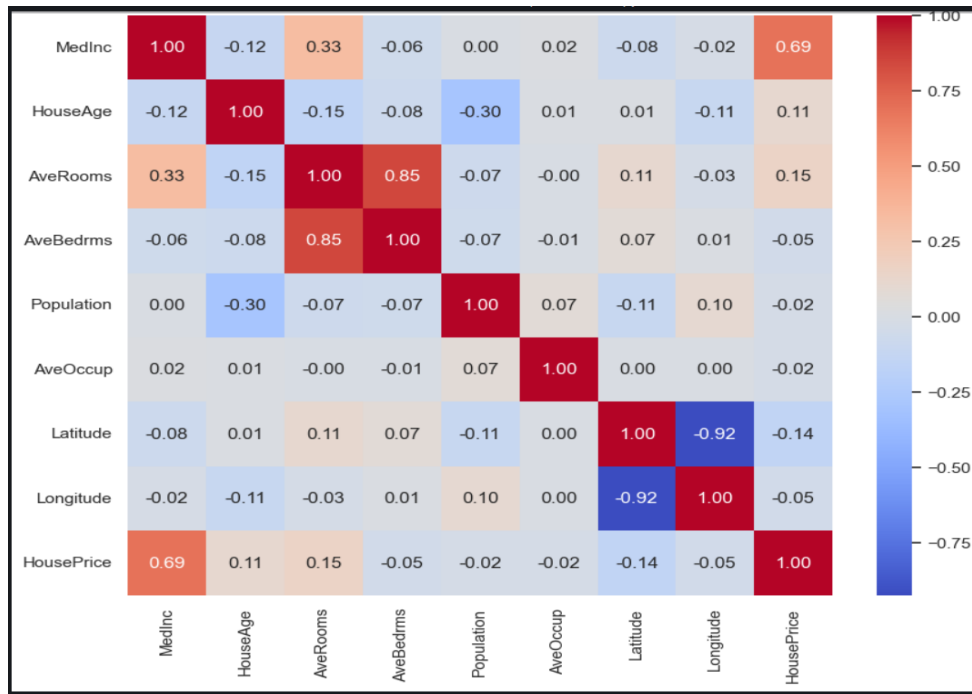
- Histograms
- Correlation Heatmap
- Scatter Plot
- Pair Plot
- Box Plot

These visualizations helped in understanding feature distributions, relationships, and possible outliers.

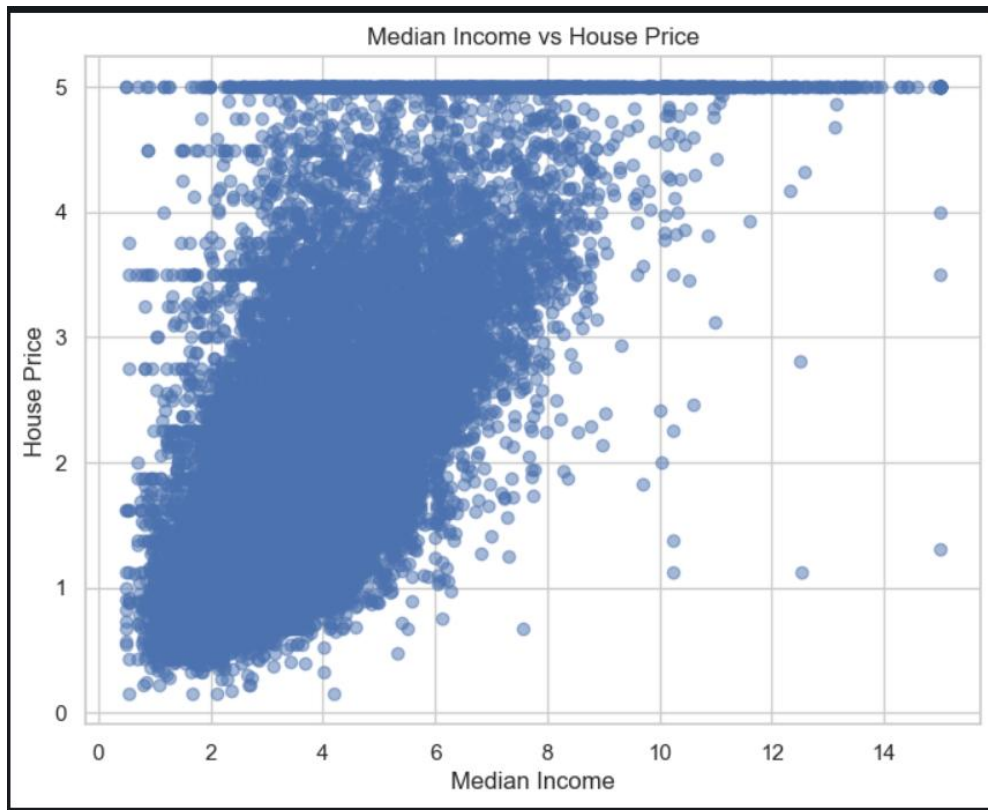
Histogram



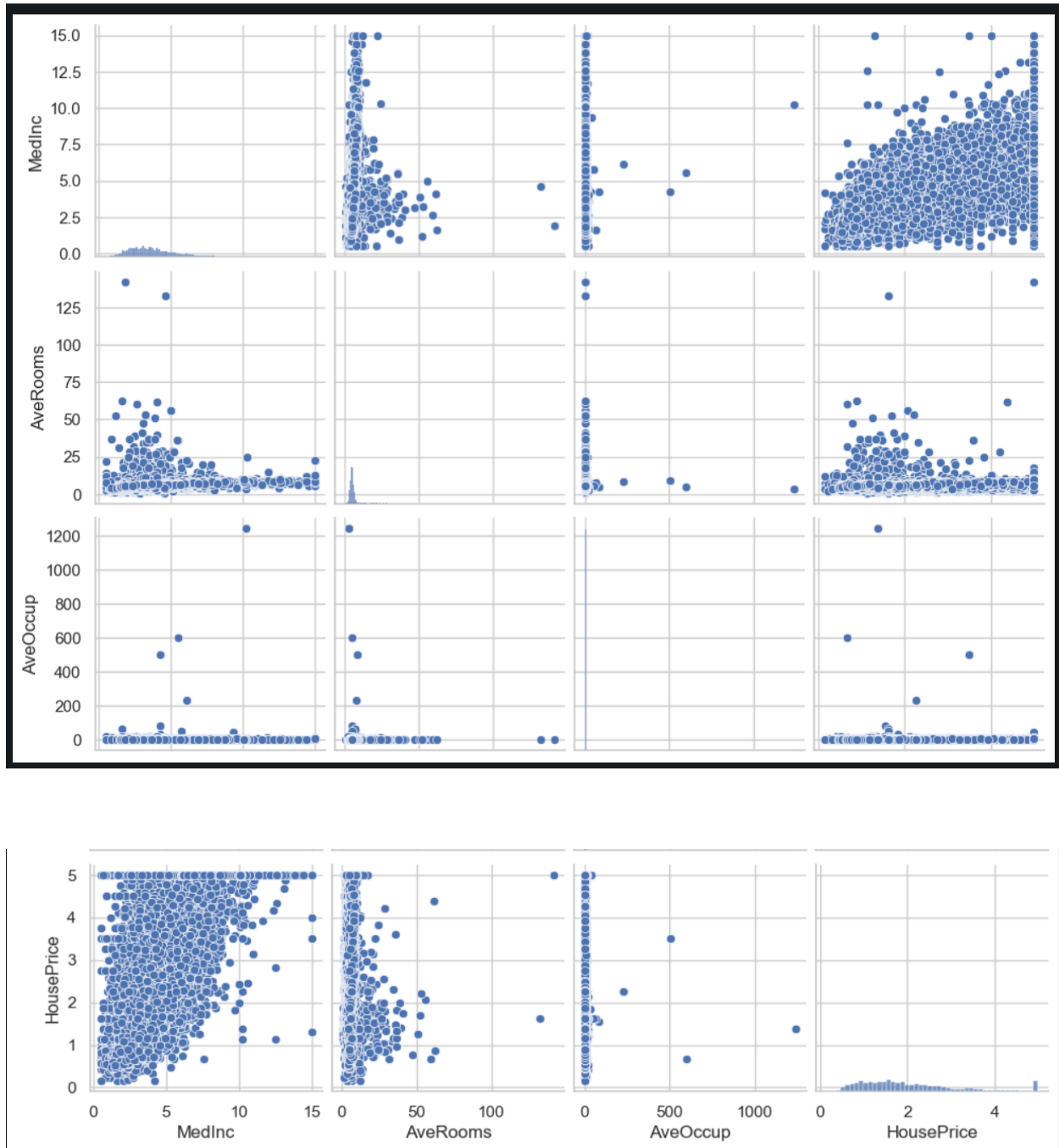
Correlation Heatmap



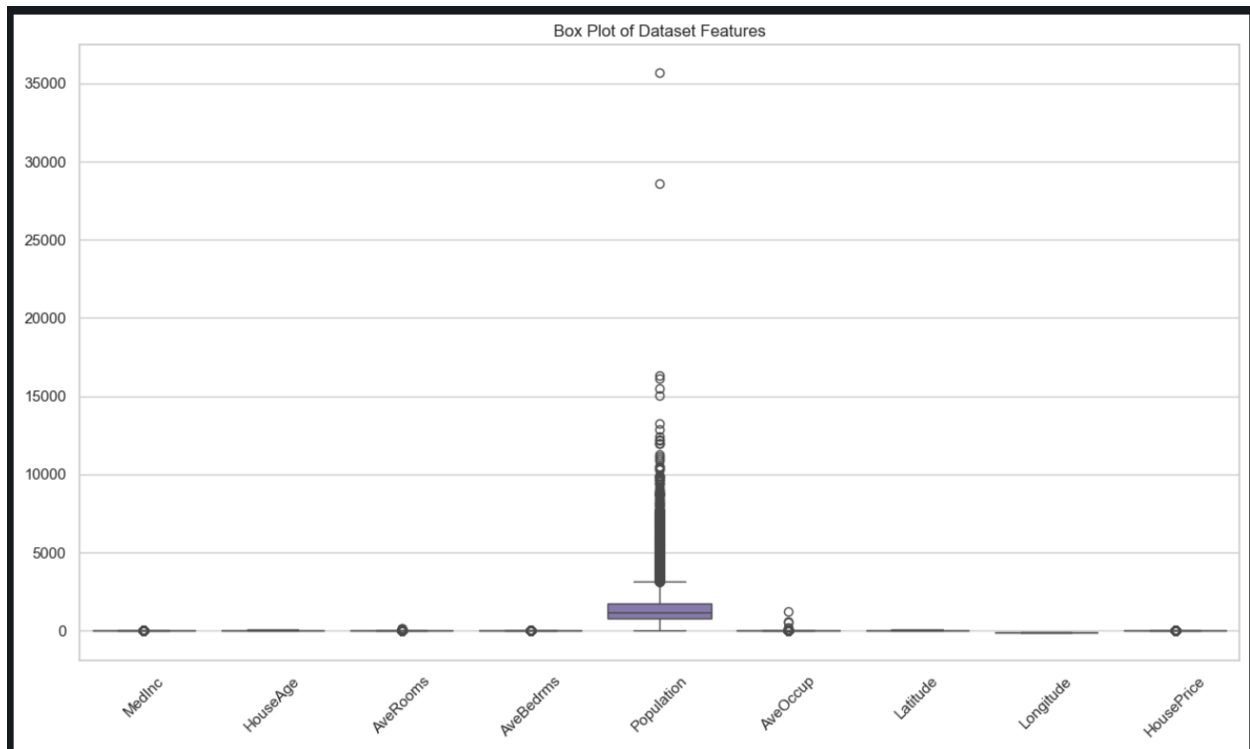
Scatter plot



Pair plot



Box plot



6. Model Training

The dataset was divided into training and testing sets using an 80:20 ratio.

A Linear Regression model was trained using the training dataset.

After training, predictions were generated for the testing dataset.

7. Model Evaluation

The model was evaluated using the following metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

- R^2 Score

Results

Metric	Value
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MAE	0.5332
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RMSE	0.7456
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R^2 Score	0.5758
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Note: Replace the placeholders with the actual values produced by your notebook.

8. Conclusion

A Linear Regression model was successfully developed using the California Housing dataset.

The project covered the complete machine learning workflow, including data exploration, visualization, preprocessing, model training, and evaluation.

The model demonstrated a reasonable ability to predict house prices using the available features.

Future improvements could include feature engineering and experimenting with more advanced machine learning algorithms to improve prediction accuracy.

9. Files Submitted

- task1_ml_linear_regression.ipynb
- house_price_model.pkl
- report.pdf
- README.md